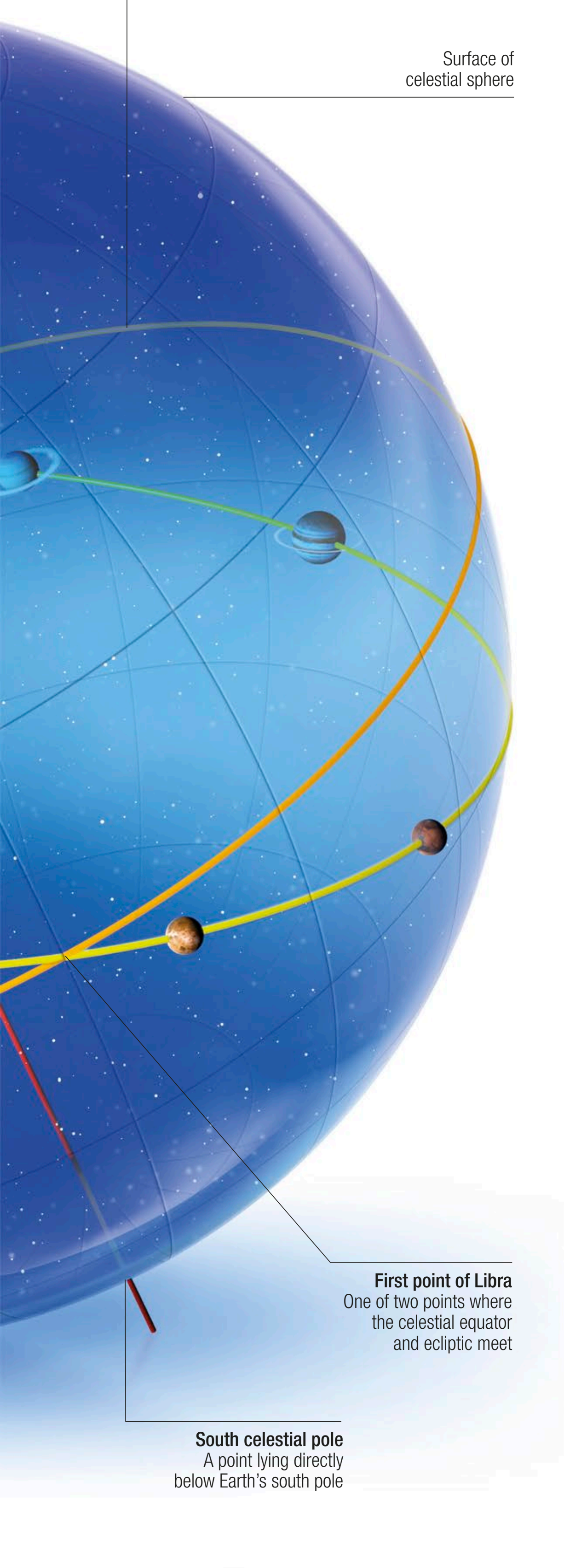
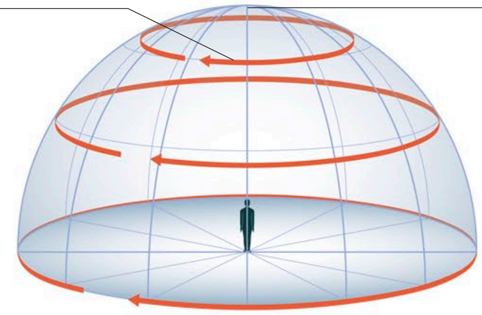
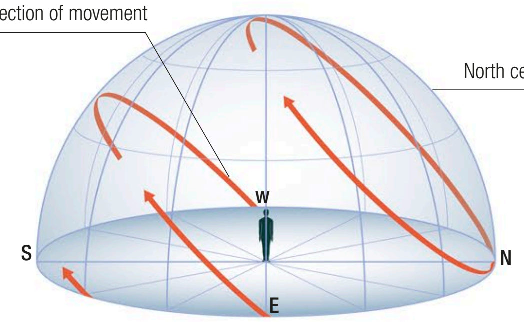


Apparent star movement

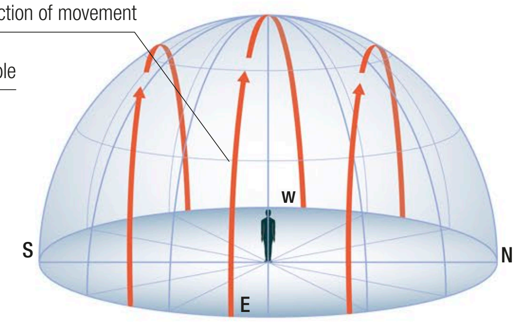
A person standing still and looking up at the night sky sees a slow, curving movement of stars and other objects across the sky. This apparent motion occurs because Earth is spinning within the celestial sphere. The pattern of motion seen varies according to the observer's location. Movements appear similar in both hemispheres, except that in the Northern Hemisphere stars appear to circle counterclockwise around the north celestial pole, while in the Southern Hemisphere they circle clockwise around the south celestial pole.



△ **Apparent motion at the North Pole**
From the observer's viewpoint, the stars seem to circle counterclockwise around a point directly overhead—the north celestial pole. Stars near the horizon move around the horizon.



△ **Apparent motion at Northern Hemisphere mid-latitudes**
For this observer, most stars rise in the east, cross the southern sky, and set in the west. But stars in the northern part of the sky circle counterclockwise around the north celestial pole.



△ **Apparent motion at the equator**
For an observer standing on or close to the equator, the stars appear to rise vertically in the east, swing overhead, and then drop vertically down again and set in the west.

Celestial coordinates

Astronomers can record the position of any object on the celestial sphere using a system of coordinates similar to that of latitude and longitude. The coordinates used by astronomers are called declination and right ascension. Declination is measured in degrees north or south of the celestial equator. Right ascension is measured in degrees east of the celestial meridian—a line that passes through both celestial poles and a point on the celestial equator called the First point of Aries.

▷ **Pinpointing a star's position**
The measurement of declination on the celestial sphere is very similar to measuring latitude on Earth's surface, while the measurement of right ascension is quite similar to the expression of longitude. The star shown here has a declination (Dec) of +45°, and a right ascension (RA) of 1 hour, or 15°.